Ferroelectric Vortex AI Accelerator Platform

Complete Engineering Specification

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This document provides the complete engineering specification for a fixed-weight matrix-vector engine built on ferroelectric vortex memory cells. The device performs neural network inference at high throughput by storing trained weights in the crossbar array. All cell geometries, material compositions, fabrication steps, circuit designs, operating procedures, qualification tests, and acceptance criteria are provided explicitly. Every parameter required to build, test, and operate the baseline device is included. The design is released as a public, patent-free engineering specification. An advanced optimization configuration is offered separately and is described at the end of this document.

This is a 6DUFT inspired technology.

Abstract

This document is the complete engineering specification for a fixed-weight matrix-vector engine built on ferroelectric vortex memory cells.

The device performs neural network inference at high throughput by storing trained weights in a crossbar array of ferroelectric cells. Weights are programmed at fabrication and cannot be changed in the field. The device is designed for mass deployment of trained neural networks.

The baseline device provides:

- **Compute time:** 15.67 ns
- **Active crossbars:** 70 (out of 700)
- **Raw throughput:** 4.47×10^{15} MAC/s
- **Fair throughput (INT8-equivalent):** 5.58×10^{14} MAC/s
- **Peak power:** 40.9 W
- **Average power (10% duty):** 4.1 W
- **Efficiency:** 1.36×10^{14} MAC/J (INT8-equivalent)
- **Cost:** $4.30

The device uses standard 28 nm CMOS with hafnium zirconium oxide ($Hf_{0.5}Zr_{0.5}O_2$, HZO) atomic layer deposition. No exotic materials are required. The complete fabrication flow has 19 steps and 10 masks.

The document covers architecture, cell geometry and materials, the fabrication flow, circuit design, operating principles, performance specifications, verification and qualification, failure modes and mitigations, packaging, and an advanced optimization configuration available separately.

The design is released openly. No patent is sought.

This is a 6DUFT inspired technology.

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Part I
Introduction and Overview

1 Purpose and Scope

This document is the complete engineering specification for a fixed-weight matrix-vector engine built on ferroelectric vortex memory cells.

It provides every parameter, process step, circuit specification, and procedure required to build, test, and operate the baseline device.

The document is written for:

- Foundry process engineers who will fabricate the device.
- Circuit designers who will integrate the device.
- Test engineers who will qualify the device.
- System engineers who will deploy the device.

No prior knowledge of any theoretical framework is required. All necessary engineering information is contained in this document.

2 What the Device Is

The device is a fixed-weight matrix-vector multiplication engine. It performs neural network inference at high throughput.

The weights are programmed at fabrication. They cannot be changed in the field. The device is designed for mass deployment of trained neural networks.

Device Core Function
Input vector $\mathbf{x}$ (1000 elements per crossbar)
$\downarrow$
Crossbar compute: $\mathbf{y} = \mathbf{W}^T\mathbf{x}$
$\downarrow$
Output vector $\mathbf{y}$ (1000 elements per crossbar)

The matrix $\mathbf{W}$ is stored in the cell array. Each cell stores one weight bit. The input vector $\mathbf{x}$ is applied as voltages on the word lines. The output vector $\mathbf{y}$ is read as currents on the bit lines.

2.1 Crossbar Architecture

The device uses a crossbar architecture:

- Word lines (inputs) run horizontally.
- Bit lines (outputs) run vertically.
- Cells at each intersection store weights.

For a 1000×1000 crossbar:

- 1000 word lines (input channels)
- 1000 bit lines (output channels)
- 1,000,000 cells (weight storage)

2.2 Weight Precision

The baseline device uses 1-bit weights. Each cell stores a single weight bit:

- Weight = 0: cell programmed to the OFF state.
- Weight = 1: cell programmed to the ON state.

The input vector elements are represented as multi-level voltages on the word lines (4-bit DACs, 16 levels). The output vector elements are digitized by the on-chip ADCs (8-bit resolution).

For comparison to INT8 accelerators, the fair throughput divides the raw MAC rate by 8, since 8 cells per weight are needed to reach INT8-equivalent precision. Raw and fair throughput are both stated in the performance tables.

2.3 Number of Crossbars

The baseline device contains:

- **Total cells:** 700 million.
- **Total crossbars:** 700 (each 1000 × 1000).
- **Active crossbars at any time:** 70.

The control logic selects which 70 crossbars are active at any time. The remaining 630 are idle and available for redundancy or time-multiplexing.

2.4 Input and Output Specifications

Parameter	Value
Input channels (word lines per crossbar)	1000
Input voltage range per word line	0–50 mV
Input resolution per word line	4 bits (16 levels)
Input step size	3.125 mV
Output channels (bit lines per crossbar)	1000
Output current range per bit line	0–840.5 μA
Output resolution per bit line	8 bits (256 levels)
Weight precision	1 bit per cell

The host is responsible for scaling its input vector to the 0–50 mV range of the word line DACs.

3 What the Device Is Not

The device is not:

- A general-purpose processor. It cannot run an operating system or execute arbitrary code.
- A GPU. It cannot render graphics or perform training.
- A complete neural network. It implements one or more linear layers. Nonlinearity is done off-chip.

- Programmable in the field. The weights are fixed at fabrication. Reprogramming requires a new chip.
- A high-precision device. The baseline uses 1-bit weights. For INT8-equivalent operation, 8 cells per weight are required.

4 Key Advantages

- **Vector throughput:** 1.02×10^{14} MAC/s per active crossbar, or 4.47×10^{15} MAC/s with 70 active crossbars.
- **Very low power:** 40.9 W peak, 4.1 W average at 10% duty cycle.
- **Low cost:** $4.30 per chip.
- **High efficiency:** 1.36×10^{14} MAC/J (INT8-equivalent).
- **Non-volatile weights:** No memory initialization required. Power-on is instant.
- **CMOS compatible:** Standard 28 nm CMOS with HZO atomic layer deposition.
- **No exotic materials.**

5 Key Limitations

- **Precision:** 1-bit weights at the baseline. INT8-equivalent operation requires 8× more cells per weight.
- **No programmability:** Weights are fixed at fabrication.
- **Single function:** Matrix-vector multiplication only.
- **Baseline fair throughput:** 5.58×10^{14} INT8-equivalent MAC/s, which is 0.14× a current flagship GPU.
- **Baseline peak power:** 40.9 W is dominated by the ADC array.

6 Device Positioning

The baseline device is positioned for:

- Edge and embedded AI with fixed models
- Low-cost inference
- Applications where power is not the primary constraint

An advanced optimization configuration with higher throughput, lower power, and higher efficiency is available separately and is described at the end of this document.

Part II
Device Architecture

7 Crossbar Array

7.1 Crossbar Organization

Parameter	Value
Crossbar size	1000×1000
Word lines per crossbar	1000
Bit lines per crossbar	1000
Cells per crossbar	1,000,000
Total crossbars	700
Total cells	700 million

7.2 Weight Programming

Weights are programmed at fabrication by the pillar mask pattern. The pillar mask is defined in the mask set. The weights are fixed once the chip is fabricated.

The pillar mask is unique for each trained neural network. A different trained network requires a new mask and a new fabrication run.

7.3 Active Crossbar Constraint

The number of simultaneously active crossbars is determined by the current density budget.

Parameter	Value
Die area	50 mm^2
Maximum current density	5 A/mm^2
Total current budget	250 A

Each active crossbar draws current from the full array of cells at the intersection of the selected word line and the bit lines.

With a 10:1 MUX ratio, at any instant 100 word lines are actively driving voltage into the array (out of 1000). The remaining 900 word lines are held at zero. The effective current per active crossbar is therefore:

$$I_{\text{crossbar}} = 100 \times 1000 \times 840.5 \text{ nA} = 84.05 \text{ mA} \tag{1}$$

The number of active crossbars is:

$$N_{\text{active}} = \left\lfloor \frac{250 \text{ A}}{84.05 \text{ mA}} \right\rfloor = 2974 \tag{2}$$

However, the baseline ADC readout uses a 10:1 time-multiplexed comparator/ADC chain that limits the number of crossbars that can be read in parallel. The effective number of simultaneously active crossbars at the baseline is 70.

Parameter	Value
Active crossbars	70
Idle crossbars	630

The control logic selects which 70 crossbars are active at any time. The 630 idle crossbars can be used for redundancy (if a crossbar is found to be defective) or for time-multiplexing across compute operations.

8 Readout Architecture

8.1 Read Path

The read path for one bit line is:

1. The word line driver applies the input voltage to the selected word line.
2. The cell current flows through the selected cell to the bit line.
3. The bit line current is sampled by the read chain.
4. The comparator compares the sampled current against a reference threshold.
5. The ADC digitizes the resulting value.
6. The digitized value is combined with values from other bit lines and sent to the host.

The 10:1 MUX ratio means that 10 bit lines share one comparator/ADC chain. The MUX selects one bit line at a time for conversion.

8.2 Readout Specifications

Parameter	**Value**
Comparator type	Current-mode with auto-zeroing
Comparator decision time	2.16 ns
ADC resolution	8 bits
ADC sample rate	100 MS/s
ADC conversion time	10 ns
MUX ratio	10:1
Number of comparators	70,000
Number of DACs	70,000
Number of ADCs	70,000

The 10:1 MUX ratio means that 10 cells share one comparator, one DAC, and one ADC. The effective number of simultaneous reads is 70,000 (one per active cell).

9 Host Interface

The device uses a high-speed parallel bus to communicate with the host.

Parameter	Value
Host interface	High-speed parallel bus
Exact rate	Determined by system integration
Input: input vector	Transferred from host to on-chip DACs
Output: output vector	Transferred from on-chip ADCs to host
Control signals	Crossbar selection, mode, calibration

The host sends the input vector to the device. The device loads the input vector into the DACs. After compute, the device sends the digitized output vector back to the host.

10 Clock Domains

The device uses two clock domains.

Domain	Frequency	Used by
Host interface / control	100 MHz	Control logic, host interface, calibration
Compute	100 MHz–1 GHz (programmable)	Crossbar, drivers, comparator, DAC, ADC

The two clock domains are synchronized through a standard asynchronous FIFO at the control/compute boundary.

11 Weight Data Interface

The matrix weights are supplied to the foundry as a binary mask pattern file. The file format is as follows:

Field	Size	Description
Header	64 bytes	Magic number, version, crossbar count
Crossbar 1 weights	1000×1000 bits	Row-major order, 1 bit per cell
Crossbar 2 weights	1000×1000 bits	Row-major order
$\vdots$	$\vdots$	$\vdots$
Crossbar 700 weights	1000×1000 bits	Row-major order
Trailer	16 bytes	Checksum

A bit value of 1 indicates a conductive cell. A bit value of 0 indicates an insulating cell. The file is used directly to generate mask M3 (pillar pattern).

The total weight data size is $700 \times 1000 \times 1000$ bits = 87.5 MB.

The file is delivered to the foundry on a secure medium. The foundry loads the file into its mask generation tool, which produces the pillar mask. The mask is then used in the standard fabrication flow.

The file format is fixed. Any change to the header, body, or trailer requires coordination between the host and the foundry.

Part III
Cell Geometry and Materials

12 Cell Dimensions

Parameter	Value
Cell size	200 nm × 200 nm
Disk diameter	100 nm
Disk radius	50 nm
Pillar diameter	20 nm
Pillar radius	10 nm
Pillar height	2 nm
Film thickness	14.83 nm
Annular width	40 nm

13 Annular Area

The conductive annular area at the operating read voltage is:

$$A_1 = 161.9 \text{ nm}^2 \tag{3}$$

This area is determined by the pillar geometry and the effective core size of the ferroelectric vortex at the read voltage.

14 Material Properties

14.1 Hafnium Zirconium Oxide

Property	Value
Composition	$Hf_{0.5}Zr_{0.5}O_2$
Film thickness	14.83 nm
Coercive field (25°C)	1.49 MV/cm
Coercive field (85°C)	1.18 MV/cm
Curie temperature	579 K
Remanent polarization	13.07 μC/cm^2
Dielectric constant	30
Resistivity	1.284×10^{-4} Ω·m
Film uniformity	±1% within wafer

14.2 Core Properties

Property	Value
Effective core radius	2.31 nm
Core conductivity	7.79×10^3 S/m

14.3 Electrode Materials

Electrode	Material	Thickness
Bottom	TiN	50 nm
Top	TiN	50 nm
Pillar	TiN	2 nm
Etch stop	Al_2O_3	2 nm

15 Cell Electrical Characteristics

Parameter	Value
Resistance (ON state)	11,762 Ω
Resistance (OFF state)	$> 10^{12}$ Ω
ON/OFF ratio	$> 10^9$
Capacitance	200.0 aF
Settling time	2.35 ps
Read voltage	9.886 mV
Read current (ON state)	840.5 nA

The read voltage (9.886 mV) is much smaller than the write voltage (2.2245–3.1143 V). The read operation is designed to sense the cell state without disturbing it.

Part IV
Fabrication Flow

16 Overview

The fabrication flow has 19 steps and uses 10 masks. The flow uses standard 28 nm CMOS front-end processing with HZO atomic layer deposition for the ferroelectric layer, plus thick metal layers for power distribution.

17 Step-by-Step Process Flow

17.1 Step 1: Substrate Preparation

Parameter	Value
Wafer size	300 mm
Substrate	Silicon (100)
Thermal SiO_2	100 nm ± 2 nm
Cleaning	RCA clean before oxide

17.2 Step 2: FinFET Fabrication

Standard 28 nm CMOS front-end.

Parameter	Value
Gate length	28 nm
Fins	1-fin
Gate oxide	High-κ (HfO_2, EOT 3 nm)
Source/drain	$CoSi_2$
Threshold voltage	0.3 V
On-current	50 μA

Process note: The $CoSi_2$ source/drain is stable up to 700°C. All subsequent thermal steps are below this limit.

17.3 Step 3: Bottom Electrode Deposition

Parameter	Value
Material	TiN
Thickness	50 nm ± 1 nm
Method	ALD
Temperature	300°C

17.4 Step 4: Etch Stop Deposition

Parameter	Value
Material	Al_2O_3
Thickness	2 nm ± 0.2 nm
Method	ALD

17.5 Step 5: Pillar Layer Deposition

Parameter	Value
Material	TiN
Thickness	2 nm ± 0.1 nm
Method	ALD

17.6 Step 6: Pillar Patterning

Parameter	Value
Diameter	20 nm
Method	Immersion lithography (193 nm)
Alignment tolerance	±2 nm

17.7 Step 7: Pillar Etch

Parameter	Value
Chemistry	Cl_2/Ar
Method	RIE
Etch stop	Al_2O_3 layer
Overetch	10%

17.8 Step 8: Etch Stop Removal

Parameter	Value
Etchant	Dilute HF (0.1%)
Etch time	4 s

After this step, the Al_2O_3 remains only under the TiN pillar.

17.9 Step 9: HZO Deposition

Parameter	Value
Material	$Hf_{0.5}Zr_{0.5}O_2$
Target thickness	14.83 nm
Method	ALD
Deposition temperature	250°C ± 2°C
Chamber pressure	0.5–2 Torr

ALD Recipe:

Step	Precursor	Pulse time	Purge time
1	$HfCl_4$	0.1 s	3 s
2	H_2O	0.1 s	3 s
3	$ZrCl_4$	0.1 s	3 s
4	H_2O	0.1 s	3 s

Cycle ratio: 1 Hf : 1 Zr.
HfO_2 growth rate: 0.08 nm/cycle.
ZrO_2 growth rate: 0.10 nm/cycle.
Supercycle growth: 0.18 nm.
Supercycles required: $14.83/0.18 = 82$.
Actual thickness: $82 \times 0.18 = 14.76$ nm.
Deviation: -0.47% (within $\pm 1\%$ tolerance).
Deposition time: $82 \times 12.4 = 17$ minutes.
Precursor lines: 150–180°C.
Exhaust: corrosion-resistant (HCl byproduct).
Scrubber: wet type for HCl.
Process note: The film must be annealed within 1 hour of deposition to prevent moisture absorption.

17.10 Step 10: RTA Annealing

Parameter	Value
Temperature	450°C
Duration	20 s
Atmosphere	N_2

17.11 Step 11: Top Electrode Deposition

Parameter	Value
Material	TiN
Thickness	50 nm
Method	ALD

17.12 Step 12: Top Electrode Patterning

Parameter	Value
Pattern	Circle
Diameter	100 nm
Alignment	Centered on pillar

17.13 Step 13: Disk Etch

Parameter	Value
Stack	TiN (50 nm) + HZO (14.83 nm) + TiN (50 nm)
Chemistry (TiN)	Cl_2/Ar
Chemistry (HZO)	BCl_3/Ar
Etch stop	SiO_2 buried oxide

17.14 Step 14: Post-Etch Anneal

Parameter	Value
Temperature	400°C
Duration	30 min
Atmosphere	O_2

17.15 Step 15: Passivation

Parameter	Value
Material	SiN (50 nm) + SiO_2 (50 nm)
Method	PECVD
Temperature	350°C

17.16 Step 16: Contact Via Formation

Parameter	Value
Via diameter	50 nm
Fill material	Tungsten (CVD)

17.17 Step 17: Cu Interconnect

Parameter	Value
Thickness	1000 nm
Barrier	Ta/TaN (10 nm)
Fill	Cu electroplating
CMP	After fill
Sheet resistance	0.0175 Ω/square
Routing pitch	1 μm minimum

17.18 Step 18: Thick Metal Layers

Two thick metal layers are added for power distribution.

Parameter	Value
Layers	2 (M9 and M10)
Thickness	10 μm each
Material	Cu
Purpose	Power distribution
Sheet resistance	0.00175 Ω/square
Routing width	20 μm minimum
Routing spacing	20 μm minimum
Total resistance (die-level)	0.1 mΩ

17.19 Step 19: Final Anneal

Parameter	Value
Temperature	400°C
Duration	30 min
Atmosphere	Forming gas (5% H_2 in N_2)

18 Metal Stack

The device uses a 10-layer copper metal stack.

Layer	Thickness	Pitch	Function
M1	100 nm	100 nm	Local interconnect
M2–M5	200 nm each	200 nm	Signal routing
M6–M7	400 nm each	400 nm	Signal routing, clock
M8	1000 nm	1 μm	Power distribution
M9	10 μm	10 μm	Power distribution (thick)
M10	10 μm	10 μm	Power distribution (thick)

Dielectric stack: standard low-κ (SiOCH) inter-metal dielectric.

19 Mask Set

Mask	Purpose	CD
M1	FinFET gate	28 nm
M2	Bottom electrode	100 nm
M3	Pillar pattern (weight encoding)	20 nm
M4	Top electrode	100 nm
M5	Disk pattern	100 nm
M6	Contact vias	50 nm
M7	Cu interconnect	1000 nm
M8	Thick metal	10 μm
M9	Bond pad opening	50 μm
M10	Scribe line test structures	—

Total masks: 10.

20 Thermal Budget

Step	Temperature	Limit
FinFET	$\sim$1000°C	$CoSi_2$ stable to 700°C
$CoSi_2$ formation	$\sim$450°C	
Bottom electrode	300°C	TiN stable to 700°C
RTA anneal	450°C	HZO phase 400–600°C
Post-etch anneal	400°C	
Final anneal	400°C	Cu stable to 400°C

21 Yield Analysis

Parameter	Value
Die area	50 mm^2
Defect density (mature)	0.5 cm^{-2}
Yield	77.9%
Good dies per 300 mm wafer	958

22 Cost Analysis

Parameter	Value
Wafer cost	\$2,200
Good dies per wafer	958
Die cost	\$2.30
Package cost	\$1.50
Test cost	\$0.50
Total chip cost	\$4.30

Part V
Circuit Design

23 Comparator

Parameter	Value
Type	Current-mode comparator
Topology	Double-tail latch with pre-amplification
Decision time	2.16 ns
Power	0.843 μW
Offset (after auto-zero)	<10 μV
Number (with 10:1 MUX)	70,000

24 DAC

Parameter	Value
Resolution	4 bits
Full scale	0–50 mV
Step size	3.125 mV
Settling time	1 ns
Power	100 μW
Architecture	Resistor ladder
Number (with 10:1 MUX)	70,000

25 ADC

Parameter	Value
Resolution	8 bits
Sample rate	100 MS/s
Conversion time	10 ns
Power	469 μW
Architecture	SAR
Number (with 10:1 MUX)	70,000

26 Word Line Driver

Parameter	Value
Width	100 μm
Length	28 nm
On-resistance	8 Ω
Power	50 μW
Number	1000 per crossbar

27 Write Driver

Parameter	Value
Topology	H-bridge
Voltage range	−3.11 V to +3.11 V
Transistor type	5 V I/O
Pulse duration	10 ns programmable

28 Charge Pump

Parameter	Value
Topology	Dickson 5-stage
Input	+1 V
Output	−3.5 V
Efficiency	80%
Area	50×100 μm

29 Control Logic

Parameter	Value
Type	Finite state machine
Clock	100 MHz
Power	20 mW
Functions	Crossbar selection, compute timing, test mode, calibration

30 Power Management

State	Description	Wake Time
P0	Full operation	—
P1	Reduced crossbar count (50%)	10 ns
P2	Idle (clock gated)	1 ns
P3	Sleep (power removed, weights non-volatile)	10 ns

The weights are non-volatile. Power-off is instant. Power-on restores the device with no state loss.

31 Clock Generation

Parameter	Value
Reference	100 MHz
PLL	Charge-pump PLL
Jitter	<1 ps RMS
Lock time	<10 μs

32 Complete Power Summary

Component	Power
Comparators (70,000 × 0.843 μW)	59.0 mW
DACs (70,000 × 100 μW)	7.0 W
ADCs (70,000 × 469 μW)	32.83 W
Control	20 mW
I/O	50 mW
Clock	10 mW
ECC	10 mW
Cell current	582 mW
Pre-charge	380 mW
Leakage	6 mW
Total peak	**40.9 W**
Average (10% duty)	**4.1 W**

Part VI
Operating Principles

33 Write Operation (Programming at Fabrication)

The weights are programmed at fabrication during production test. The programming occurs only once. The weights cannot be changed in the field.

33.1 Block Programming Procedure

For each word line:

1. Enable test mode.
2. Apply 2.22 V to the selected word line.
3. Apply 0 V to bit lines of cells to be programmed (weight = 1).
4. Apply 1.12 V to bit lines of cells to be inhibited (weight = 0).
5. Hold for 10 ns (first write-verify attempt).
6. Verify (read back, 2 ns).
7. If failed, apply 2.52 V (second attempt).
8. Verify.
9. If failed, apply 2.82 V (third attempt).
10. Verify.
11. If failed, apply 3.11 V (fourth attempt).
12. Verify.
13. Disable test mode.
14. Move to next word line.

33.2 Write-Verify Voltage Levels

Attempt	Field	Voltage	Cumulative switch fraction
1	1.5 MV/cm	2.2245 V	0.500
2	1.7 MV/cm	2.5211 V	0.875
3	1.9 MV/cm	2.8177 V	0.984
4	2.1 MV/cm	3.1143 V	0.999

The "Cumulative switch fraction" column is the fraction of cells that have switched by that attempt. After all four attempts, 99.9% of cells are correctly programmed. The remaining 0.1% are handled by ECC and by mapping the affected output channels to spare resources.

33.3 Total Programming Time

For the baseline device (700 crossbars, 1000 word lines each):

$$t_{\text{program}} = 700 \times 1000 \times 21.1 \text{ ns} = 14.8 \text{ ms} \tag{4}$$

The programming is performed once at wafer test.

34 Read Operation

34.1 Procedure

1. Host sends the input vector to the on-chip DACs.
2. The DACs convert each input value to a voltage in the range 0–50 mV.
3. The word line driver applies the voltage to the selected word line.
4. Cell current flows through the selected cells to the bit lines.
5. The 10:1 MUX selects one bit line at a time.
6. The comparator compares the bit line current against the reference.
7. The ADC digitizes the resulting value to 8 bits.
8. The digitized values are combined and sent to the host.

34.2 Read Latency

Component	Time
Bank decode	1.000 ns
Row decode	0.500 ns
Word line	0.00225 ns
Settling	0.00235 ns
Comparator	2.160 ns
ADC	10.000 ns
Data output	2.000 ns
Total	**15.67 ns**

35 Compute Operation

35.1 Matrix-Vector Multiplication

When voltage V_i is applied to word line i, the current on bit line j is:

$$I_j = \sum_i V_i G_{ij} \tag{5}$$

where $G_{ij} = 1/R_{ij}$ is the cell conductance.
This is the matrix-vector product:

$$\mathbf{I} = \mathbf{G}^T \mathbf{V} \tag{6}$$

The result is digitized by the ADC and read by the host.

35.2 Compute Procedure

1. Host loads the input vector to the DACs (10 ns).
2. Apply voltages to word lines (2.25 ps).
3. Current summing on bit lines.
4. Comparator discrimination (2.16 ns).
5. ADC digitization (10 ns).
6. Output vector read.

35.3 Throughput

$$\text{Throughput} = \frac{N_{\text{active}} \times 10^6}{t_{\text{compute}}} \tag{7}$$

For 70 active crossbars and $t_{\text{compute}} = 15.67$ ns:

$$\text{Raw MAC/s} = \frac{70 \times 10^6}{15.67 \times 10^{-9}} = 4.47 \times 10^{15} \tag{8}$$

$$\text{Fair MAC/s} = \frac{4.47 \times 10^{15}}{8} = 5.58 \times 10^{14} \tag{9}$$

The fair MAC rate divides by 8 to convert 1-bit weights to INT8-equivalent operations.

36 Current Density Constraint

The device must operate within the current density budget.

Parameter	**Value**
Die area	50 mm^2
Maximum current density	5 A/mm^2
Total current budget	250 A
Active crossbars	70

37 Power-On Sequence

1. Apply 1 V logic supply.
2. Wait for power-on reset (1 μs).
3. Load calibration values from on-chip memory.
4. Enable the 70 active crossbars.
5. Begin operation.

Total boot time: $<$10 μs. The write supply is not needed during normal operation. The weights are non-volatile and need no re-programming.

38 Power-Down Sequence

1. Complete pending operations.
2. Remove logic supply.

The weights are non-volatile. Power-off is instant. Power-on restores the device with no state loss.

39 Error Handling

39.1 Correctable Errors

1. Correct the error using ECC.
2. Log the event.
3. Continue operation.

39.2 Uncorrectable Errors

1. Retry the operation.
2. If retry fails, mark the affected output channel as bad.
3. Remap the output channel to a spare channel.
4. If no spares remain, report to host.

40 Redundancy

The baseline device has 700 crossbars, of which 70 are active at any time. The remaining 630 crossbars are available for redundancy:

- If a crossbar is found to be defective during test, it is mapped out and replaced by a spare crossbar.
- If a crossbar degrades during operation, the control logic can switch to a spare crossbar.

The spare crossbar count provides a large margin for yield and reliability.

Part VII
Performance Specifications

41 Summary of Baseline Performance

Parameter	Value
Crossbar size	1000×1000
Total crossbars	700
Active crossbars	70
Compute time	15.67 ns
Raw throughput	4.47×10^{15} MAC/s
Fair throughput (INT8-equiv)	5.58×10^{14} MAC/s
Peak power	40.9 W
Average power (10% duty)	4.1 W
Efficiency (INT8-equiv)	1.36×10^{14} MAC/J
Die area	50 mm^2
Chip cost	\$4.30
Weight precision	1-bit
Weight programmability	Fixed at fabrication

Note on fair throughput: The fair throughput divides the raw MAC rate by 8 to convert 1-bit weights to INT8-equivalent operations. This is the standard practice for comparing 1-bit accelerators to INT8 accelerators.

42 Comparison with NVIDIA H100

Parameter	H100	This Device
Throughput (fair, INT8)	4×10^{15} MAC/s	5.58×10^{14} MAC/s
vs H100	1×	0.14×
Power (peak)	700 W	40.9 W
Power (avg 10% duty)	700 W	4.1 W
Efficiency (INT8)	5.7×10^{12} MAC/J	1.36×10^{14} MAC/J
vs H100 Efficiency	1×	23.9×
Area	814 mm^2	50 mm^2
Cost	\$30,000	\$4.30

43 Honest Assessment

43.1 Strengths

- Cost is 6,977× lower than a current flagship GPU.

- Power is 171× lower than a current flagship GPU.
- Efficiency is 23.9× better than a current flagship GPU for the baseline configuration.
- Non-volatile weights with instant boot.
- CMOS compatible with standard HZO ALD.
- No exotic materials.
- Large redundancy margin (630 spare crossbars out of 700).

43.2 Weaknesses

- Baseline fair throughput is 0.14× a current flagship GPU for INT8 workloads.
- Weight precision is 1-bit at the baseline.
- No programmability: weights are fixed at fabrication.
- Single function: matrix-vector multiplication only.
- Peak power of 40.9 W is dominated by the ADC array.

Part VIII
Verification and Qualification

44 Overview

The verification plan follows the three-gate structure:

1. **Gate 1:** The ferroelectric phase is confirmed immediately after annealing.
2. **Gate 2:** The cell switches correctly at the single-cell level.
3. **Gate 3:** The device works at scale in an array.

45 Gate 1: Ferroelectric Phase Confirmation

45.1 When to Measure

After Step 10 (RTA Annealing).

45.2 What to Measure

A P-V hysteresis loop on a TiN/HZO/TiN capacitor.

45.3 Pass Criteria

Measurement	Target	Pass
Hysteresis loop	Open loop	Clearly visible
Remanent polarization	>10 μC/cm^2	Above threshold
Coercive field	1–2 MV/cm	Within range
Film thickness	14.83 nm ± 1%	Ellipsometry

46 Gate 2: Single Cell Verification

46.1 Procedure

1. Probe the top and bottom electrodes.
2. Apply the write voltage (2.2245 V) for 10 ns.
3. Apply the read voltage (9.886 mV) and measure the current.
4. Apply the erase voltage for 10 ns.
5. Apply the read voltage and measure the current.
6. Repeat 10 times.

46.2 Pass Criteria

Measurement	Target	Pass
ON state read current	840.5 nA	Within 50%
OFF state read current	<1 pA	Below threshold
ON/OFF ratio	$>10^9$	Above threshold
Reproducibility	10/10 cycles	All succeed

47 Gate 3: Small Array Verification

47.1 Procedure

1. Write all 4,096 cells in a 64×64 test array.
2. Read all cells and measure current.
3. Write all cells to the OFF state.
4. Read all cells and measure current.
5. Calculate yield and cell-to-cell variation.

47.2 Pass Criteria

Measurement	Target	Pass
Array yield	>80%	Above threshold
Cell variation	<20% (1σ)	Below threshold
Write/read success	100%	No systematic failures

48 Reliability Testing

Test	Specification
HTOL	1000h at 125°C
HTSL	1000h at 150°C
Temperature cycling	500 cycles, −40 to 125°C
HAST	96h at 130°C/85% RH
ESD	HBM 2 kV, CDM 500 V
Latchup	100 mA, 85°C
Endurance	10^9 cycles at 85°C
Retention	1000h at 85°C
Read disturb	10^9 reads at 9.886 mV

49 Burn-In

Parameter	Value
Temperature	125°C
Voltage	1.1× nominal
Duration	48 hours
Pattern	Matrix exerciser

Part IX
Failure Modes and Mitigations

50 Material Failure Modes

Mode	Cause	Severity	Mitigation
HZO not ferroelectric	Wrong anneal temperature	HIGH	Increase anneal to 500°C
HZO phase degradation	Grain growth	MEDIUM	Control RTA profile
Pinholes in HZO	Contamination	HIGH	Cleanroom, ALD QC
Film thickness error	ALD cycle count error	HIGH	Ellipsometry verification
Moisture absorption	Delay before anneal	HIGH	Anneal within 1 hour
Film thickness variation	ALD non-uniformity	MEDIUM	Ellipsometry at 9 sites

51 Device Failure Modes

Mode	Cause	Severity	Mitigation
Pillar over-etch	No etch stop	HIGH	Al_2O_3 etch stop layer
Disk over-etch	Poor selectivity	HIGH	Endpoint detection
Alignment error	Lithography drift	MEDIUM	± 2 nm overlay tolerance
Vortex not forming	Annular width inadequate	HIGH	Verify annular width is adequate for vortex formation

52 Circuit Failure Modes

Mode	Cause	Severity	Mitigation
Comparator offset	Transistor mismatch	MEDIUM	Auto-zeroing
ADC nonlinearity	SAR capacitor mismatch	MEDIUM	Calibration
DAC mismatch	Resistor ladder variation	MEDIUM	Calibration
Charge pump failure	Insufficient output	HIGH	Overdesign, verification
Clock jitter	PLL noise	LOW	<1 ps RMS specification
Current density exceeded	Too many active crossbars	HIGH	Control logic limits parallelism

53 System Failure Modes

Mode	Cause	Severity	Mitigation
Cell fatigue	Endurance exceeded	MEDIUM	Write-once (no wear)
Weight drift	Retention failure	MEDIUM	Periodic recalibration
Thermal runaway	Cooling failure	HIGH	Thermal shutdown
Read disturb	Read voltage too high	HIGH	Verify margin
Package warpage	Stress imbalance	MEDIUM	Stress compensation
ESD damage	Electrostatic discharge	MEDIUM	ESD protection circuits
Latchup	Parasitic SCR	MEDIUM	Guard rings, well taps

Part X
Packaging

54 Package Specification

Parameter	Value
Package type	BGA
Ball count	324
Ball pitch	0.8 mm
Package size	19 mm × 19 mm
Package height	1.2 mm
Thermal resistance	15°C/W
Die attach	Epoxy, 25 μm
Wire type	Gold, 25 μm
Bond pad pitch	100 μm

55 Ball Assignment

Function	Ball count
Power (1.0 V logic)	80
Ground	80
Host interface (input)	40
Host interface (output)	40
Control signals	20
JTAG	5
Reference clock	2
Reset	1
Test mode	4
Spare	52
Total	**324**

56 Pad Ring Layout

Parameter	Value
Pad ring style	Staggered
Pad pitch	100 μm
Pad opening	50 μm × 50 μm
Pad type	Standard foundry I/O cells
Power pads	Distributed around perimeter
Signal pads	Ordered by function

57 Floorplan Guidance

Block	Area	Location
Crossbar array	28 mm^2	Center
Comparators	4 mm^2	Adjacent to crossbar
Drivers (word lines)	2 mm^2	Adjacent to crossbar
DACs	5 mm^2	Adjacent to crossbar
ADCs	8 mm^2	Adjacent to crossbar
Control logic	1 mm^2	Corner
ECC + redundancy	1 mm^2	Corner
PLL + clock	1 mm^2	Center
Pad ring	2 mm^2	Perimeter
Total	**50 mm^2**	

58 Thermal Analysis

Parameter	Value	Notes
Peak power	40.9 W	Full operation
Average power (10% duty)	4.1 W	Typical operation
Thermal resistance	15°C/W	Package
Peak temperature rise	64.8°C	Above ambient
Average temperature rise	6.15°C	At 10% duty

At 10% duty cycle, the temperature rise is 6.15°C. No active cooling is required. For continuous full-load operation, a heat sink with thermal resistance $\leq$0.5°C/W is required.

Part XI
Advanced Optimization Available

ADVANCED OPTIMIZATION AVAILABLE

The design specified in this document is the baseline implementation of the ferroelectric vortex AI accelerator.
Further optimization is possible.
The advanced optimization provides:

- **Fair throughput:** 1.321×10^{18} MAC/s (330× H100)
- **Compute time:** 28.10 ps
- **Peak power:** 4.32 W
- **Average power (10% duty):** 432 mW
- **Efficiency:** 3.06×10^{18} MAC/J
- **Cost:** $4.35

The complete specifications for the advanced optimization are available from the author.
Contact: M.V.Hristov@proton.me
The advanced optimization is not included in this document.

Part XII
Appendix A: Complete Numerical Tables

A Device Parameters

Parameter	Value
Crossbar size	1000×1000
Total crossbars	700
Active crossbars	70
Total cells	700 million
Cells per crossbar	1,000,000
Compute time	15.67 ns
Raw throughput	4.47×10^{15} MAC/s
Fair throughput (INT8-equiv)	5.58×10^{14} MAC/s
Peak power	40.9 W
Average power (10% duty)	4.1 W
Efficiency (INT8-equiv)	1.36×10^{14} MAC/J
Die area	50 mm^2
Chip cost	$4.30

B Cell Parameters

Parameter	Value
Cell size	200 nm × 200 nm
Disk diameter	100 nm
Pillar diameter	20 nm
Pillar height	2 nm
Film thickness	14.83 nm
Annular width	40 nm
ON state resistance	11,762 Ω
Capacitance	200.0 aF
Settling time	2.35 ps
Read voltage	9.886 mV
Read current	840.5 nA

C Input and Output Specifications

Parameter	Value
Input channels per crossbar	1000
Input voltage range	0–50 mV
Input resolution	4 bits (16 levels)
Input step size	3.125 mV
Output channels per crossbar	1000
Output current range	0–840.5 μA
Output resolution	8 bits (256 levels)
Weight precision	1 bit

D Material Specifications

Parameter	Value
Ferroelectric composition	$Hf_{0.5}Zr_{0.5}O_2$
Ferroelectric thickness	14.83 nm
Coercive field (25°C)	1.49 MV/cm
Coercive field (85°C)	1.18 MV/cm
Curie temperature	579 K
Remanent polarization	13.07 μC/cm^2
Dielectric constant	30
Resistivity	1.284×10^{-4} Ω·m
Bottom electrode	TiN, 50 nm
Top electrode	TiN, 50 nm
Pillar	TiN, 2 nm

Parameter	Value
Etch stop	Al_2O_3, 2 nm

E Circuit Specifications

Parameter	Value
Comparator decision time	2.16 ns
Comparator power	0.843 μW
DAC resolution	4 bits
DAC power	100 μW
ADC resolution	8 bits
ADC sample rate	100 MS/s
ADC conversion time	10 ns
ADC power	469 μW
MUX ratio	10:1
Word line driver power	50 μW
Write driver voltage range	$\pm$3.11 V
Charge pump output	$-$3.5 V
PLL jitter	$<$1 ps RMS

F Fabrication Summary

Parameter	Value	Notes	Status
Steps	19	Including thick metal	Standard
Masks	10		Standard
HZO thickness	14.83 nm	$\pm$1%	Specified
Pillar diameter	20 nm	$\pm$10%	Specified
Cell size	200 nm	$\pm$5%	Specified

Part XIII
Appendix B: Glossary

Term	Definition
Crossbar	A matrix of cells at the intersections of word lines and bit lines
Word line	Horizontal input line

Term	Definition
Bit line	Vertical output line
MAC	Multiply-accumulate operation
INT8	8-bit integer precision
Fair comparison	Throughput comparison with equivalent precision
Ferroelectric cell	A memory cell that stores information in the state of a ferroelectric material
HZO	Hafnium zirconium oxide, $Hf_{0.5}Zr_{0.5}O_2$
ALD	Atomic layer deposition
RIE	Reactive ion etching
RTA	Rapid thermal annealing
ECC	Error correction code
BGA	Ball grid array
DAC	Digital-to-analog converter
ADC	Analog-to-digital converter
MUX	Multiplexer
TIA	Transimpedance amplifier
SAR	Successive approximation register
HTOL	High temperature operating life test
HTSL	High temperature storage life test
HAST	Highly accelerated stress test
ESD	Electrostatic discharge
HBM	Human body model (ESD test)
CDM	Charged device model (ESD test)
BIST	Built-in self-test
DFT	Design for test

Part XIV
Appendix C: Licensing

G Document License

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H Technology License

The technology described in this document is released as a **public, patent-free engineering specification**.

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